



EETS 7304

Module 5 Part 1 Real-Time Protocol (RTP) Session Initiation Protocol (SIP)

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Topics/Objectives

- Voice over IP – Real-Time Protocol (RTP), Real-Time Control Protocol and Real-Time Streaming Protocol
- Session Initiation Protocol (SIP)



Real-Time Protocol (RTP)

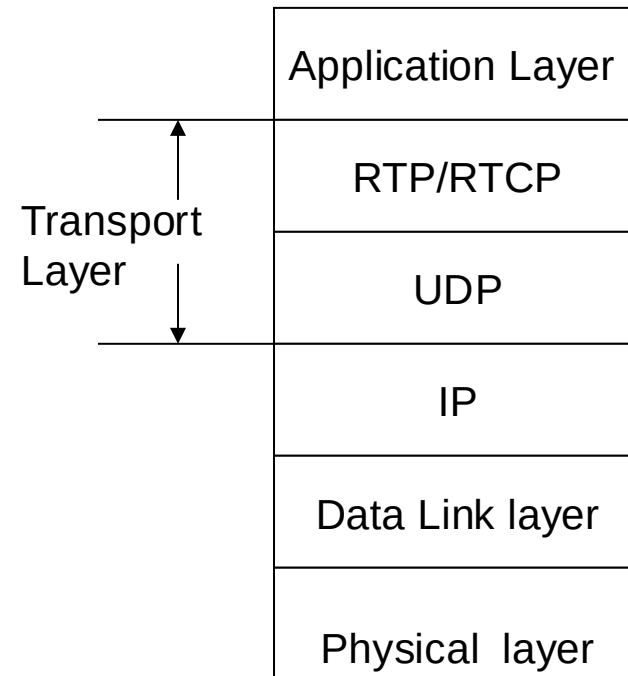


VoIP at Layer 4

- Real-Time Protocol (RTP)
 - IETF RFC 3550
- Real-Time Control Protocol (RTCP)
 - IETF RFC 3550
- Real-Time Streaming Protocol (RTSP)
 - IETF RFC 2326



RTP at the Transport Layer



Real-Time Protocol (RTP)

- A framing protocol for real-time services.
- End-to-end transport functions via unicast or multicast.
- Has provisions to help overcome jitter.
- Only used by end systems.

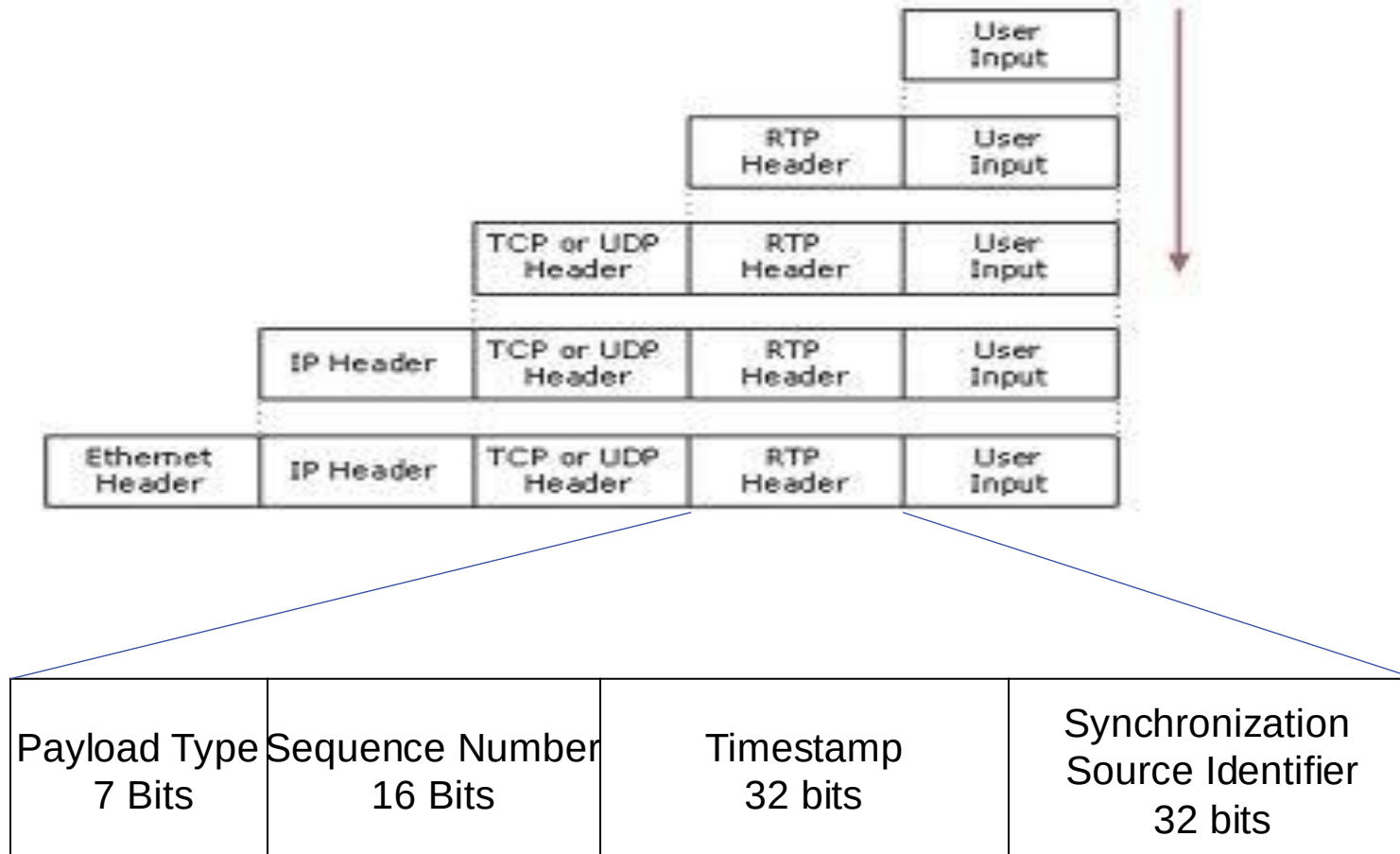


RTP does not:

- RTP itself does not provide any mechanism to ensure timely delivery of packets
- Does not guarantee delivery of packets or prevent out-of-order delivery.
- Provide QoS capability but some routers can enable priority transmission for RTP assigned packets.
 - For example, Cisco IP RTP PRIORITY command.



RTP Packet Header



RTP Packet Header

- Payload type indicates encoding used for audio and video
- Sequence Number
 - Increments by one for each RTP packet
 - Used to detect packet losses
- Time Stamp
 - Denotes the sampling instant of the first byte in the RTP data
 - To remove packet jitter
 - Derived from sampling clock at the sender
- SSRI
 - Identifies the source of the RTP stream
 - Distinct from IP address



RTP Control Protocol (RTCP)

- RTCP is used during audio/video transmission
- RTCP packets sent periodically, not constantly.
 - RTCP traffic should not exceed 5% of total session bandwidth
- RTCP packets are distributed to all the participants using IP multicast
- Distinguished from RTP through the use of distinct port numbers
- Feedback on Quality of Service via various reports:
 - Number of packets sent
 - Number of packets lost
 - Inter-arrival time jitter
- Up to the application to make use of RTCP packets



Real-Time Streaming Protocol (RTSP)

- Act as a network remote control extension.
- Supports the following operations and others:
 - Retrieval of media from a server.
 - Invitation of a media server to a conference.
 - Recording of a conference.
- Protocol design:
 - Text-based, similar to HTTP.
 - Supports and session protocol.
 - Unicast or multicast



Wireshark Links

<http://wiki.wireshark.org/SampleCaptures>

http://wiki.wireshark.org/SampleCaptures?action=AttachFile&do=view&target=SIP_CALL_RTP_G711

<http://packetlife.net/captures/protocol/ipv6/>



Session Initiation Protocol (SIP)

Components, use cases, and operations



Session Initiation Protocol (SIP)

- First described in IETF RFC 2543
- Defined the following functions to determine:
 - User location
 - User availability
 - Endpoint capabilities
 - Session setup
 - Session management
- SIP 2.0 most current, described in RFC 3261.



SIP Standards

- SIP 2.0 most current, described in RFC3261.
- Dozens of related RFCs now.
- SIP is certainly not simple anymore!
 - Not complete, still.
 - PSTN more complex than developers originally anticipated
 - Initial shunning layering/modular design was a mistake.
- Vast improvement with adoption by 3GPP/3GPP2

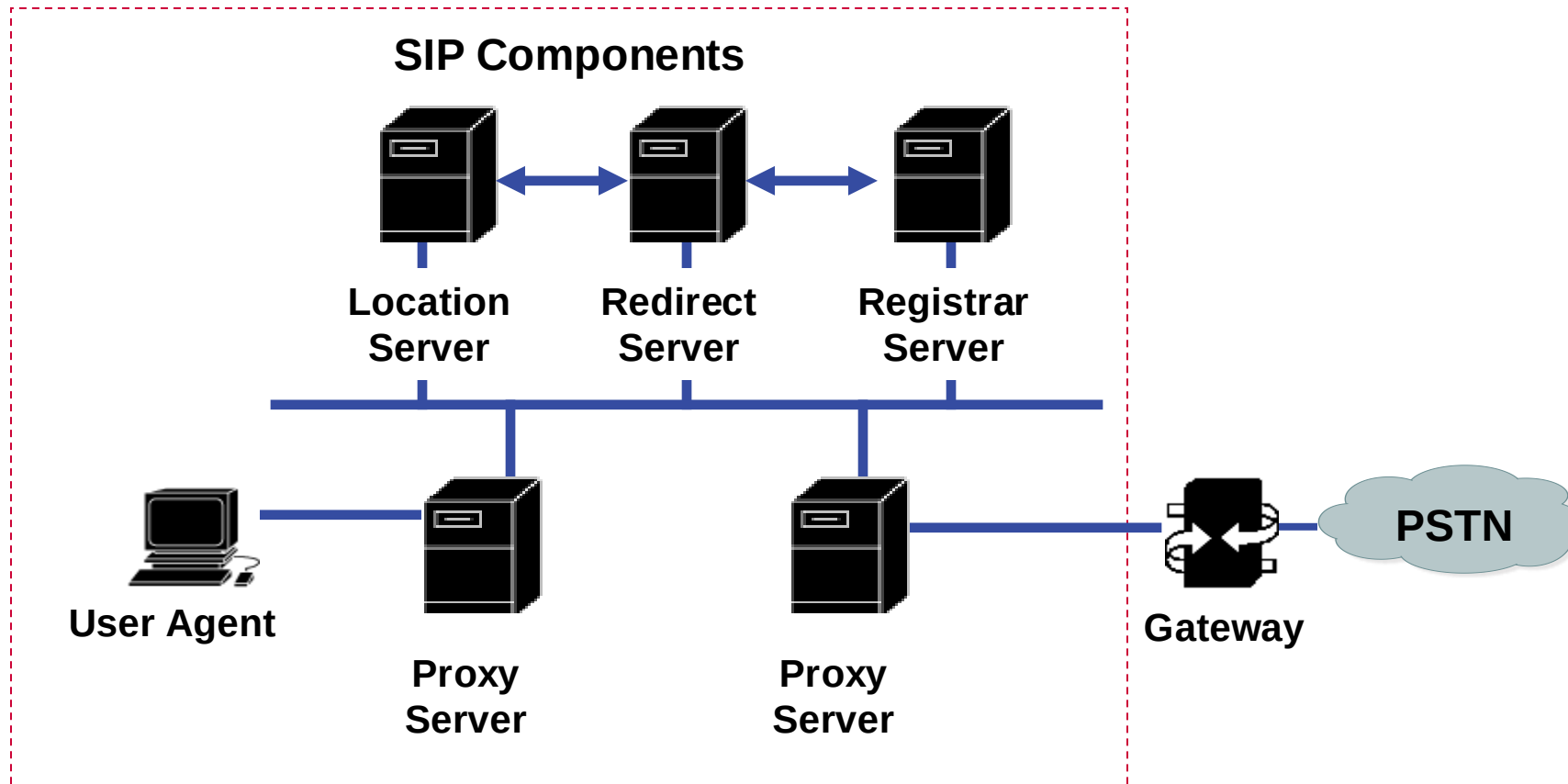


H.323 versus SIP

Information	SIP	H.323
Standards Body	IETF.	ITU.
Relationship	Peer-to-Peer.	Peer-to-Peer.
Origins	Internet based and web centric. Borrows syntax and messages from HTTP.	Telephony based. Borrows call signaling protocol from ISDN Q.SIG.
Client	Intelligent user agents.	Intelligent H.323 terminals.
Core servers	SIP proxy, redirect, location, and registration servers.	H.323 Gatekeeper.
Current Deployment	Gaining dominance.	Widespread.



SIP Distributed Architecture

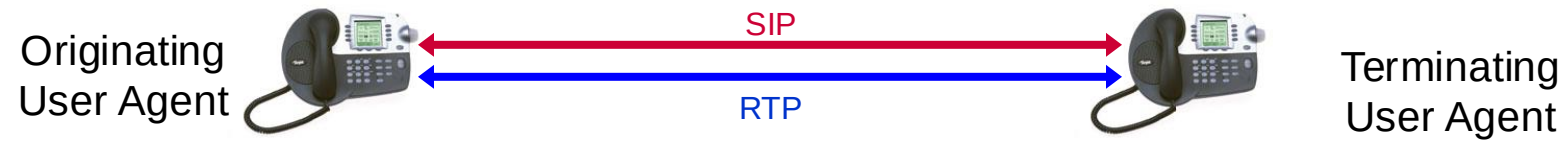


SIP User Agent (UA)

- An intermediary program that acts as both a server and a client to make requests on behalf of other clients.
- Requests are serviced internally or by passing them on, possibly after translation, to other servers.
- Interprets, rewrites or translates a request message before forwarding it.

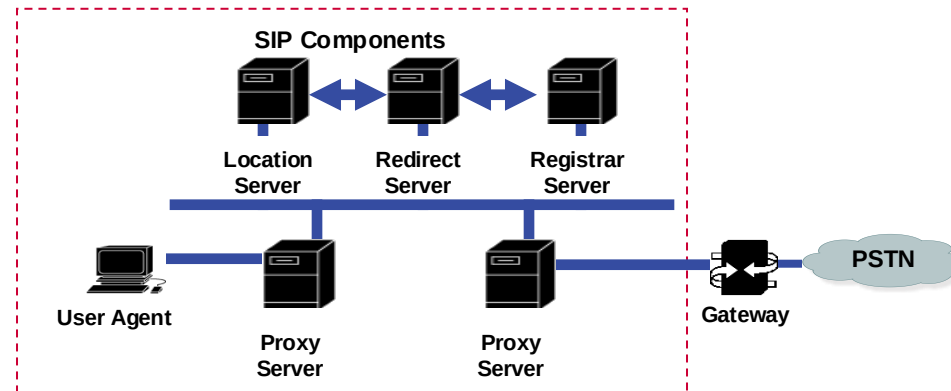


SIP UA-to-UA



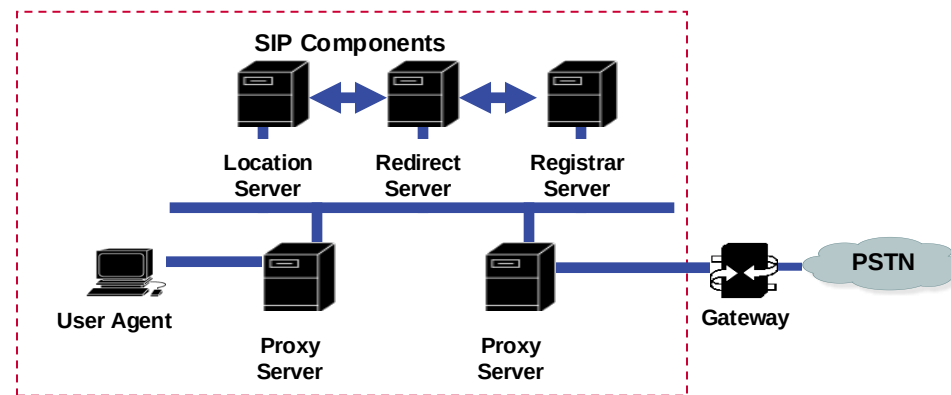
SIP Proxy Server

- An intermediary program that acts as both a server and a client to make requests on behalf of other clients.
- Requests are serviced internally or by passing them on, possibly after translation, to other servers.
- Interprets, rewrites or translates a request message before forwarding it.



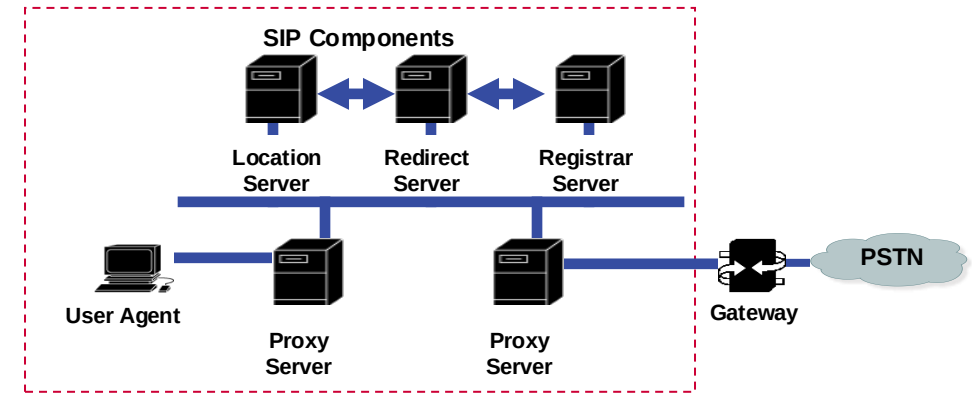
SIP Local User Proxy Server

- Primarily exist in customer IP networks
- Provides endpoint mobility services via SIP registration methods
- Provides wide range of user specific services
 - simple canned service
 - full user controlled services via CGI or CPL scripts or servlets



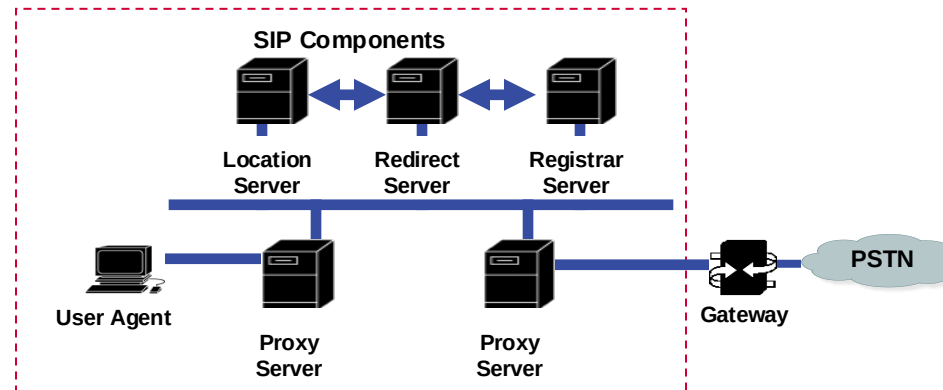
SIP Network User Proxy Server

- “Network Servers” that Proxy requests to other server.
- User Proxies manage sets of SIP users.
- Provides dynamic association of SIP endpoints.
- Proxy capabilities:
 - outgoing calls
 - address lookup
 - policy management
 - firewalls
- Proxy servers can be stateful or stateless.



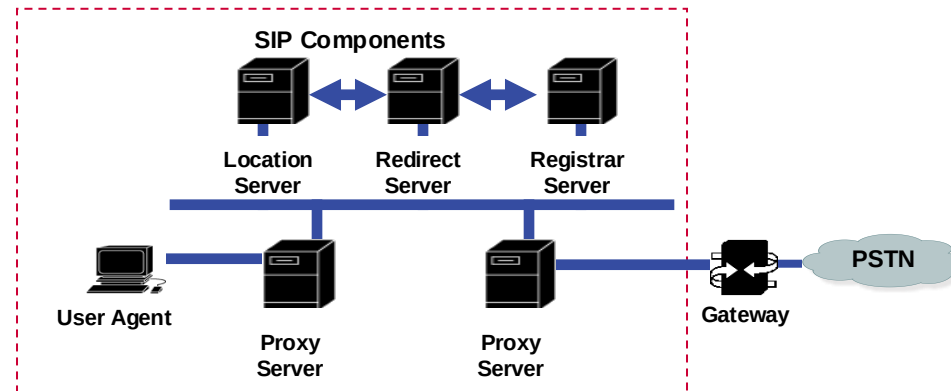
SIP Location Server

- A location server is used by a SIP redirect or proxy server to obtain information about a called party's possible location(s).



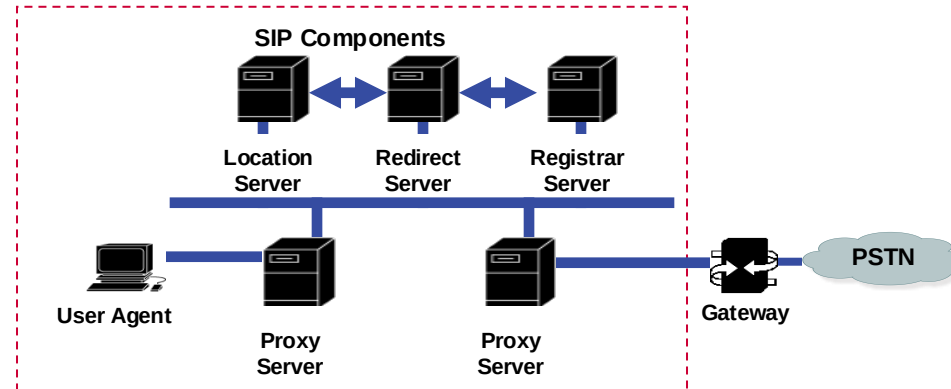
SIP Redirect Server

- A server that accepts a SIP request, maps the address into zero or more new addresses and returns these addresses to the client.
- Unlike a proxy server, the redirect server does not initiate its own SIP request.
- Unlike a user agent server, the redirect server does not accept or terminate calls.



SIP Registrar Server

- A server that accepts REGISTER requests.
- The register server may support authentication.
- A registrar server is typically co-located with a proxy or redirect server and may offer location services.

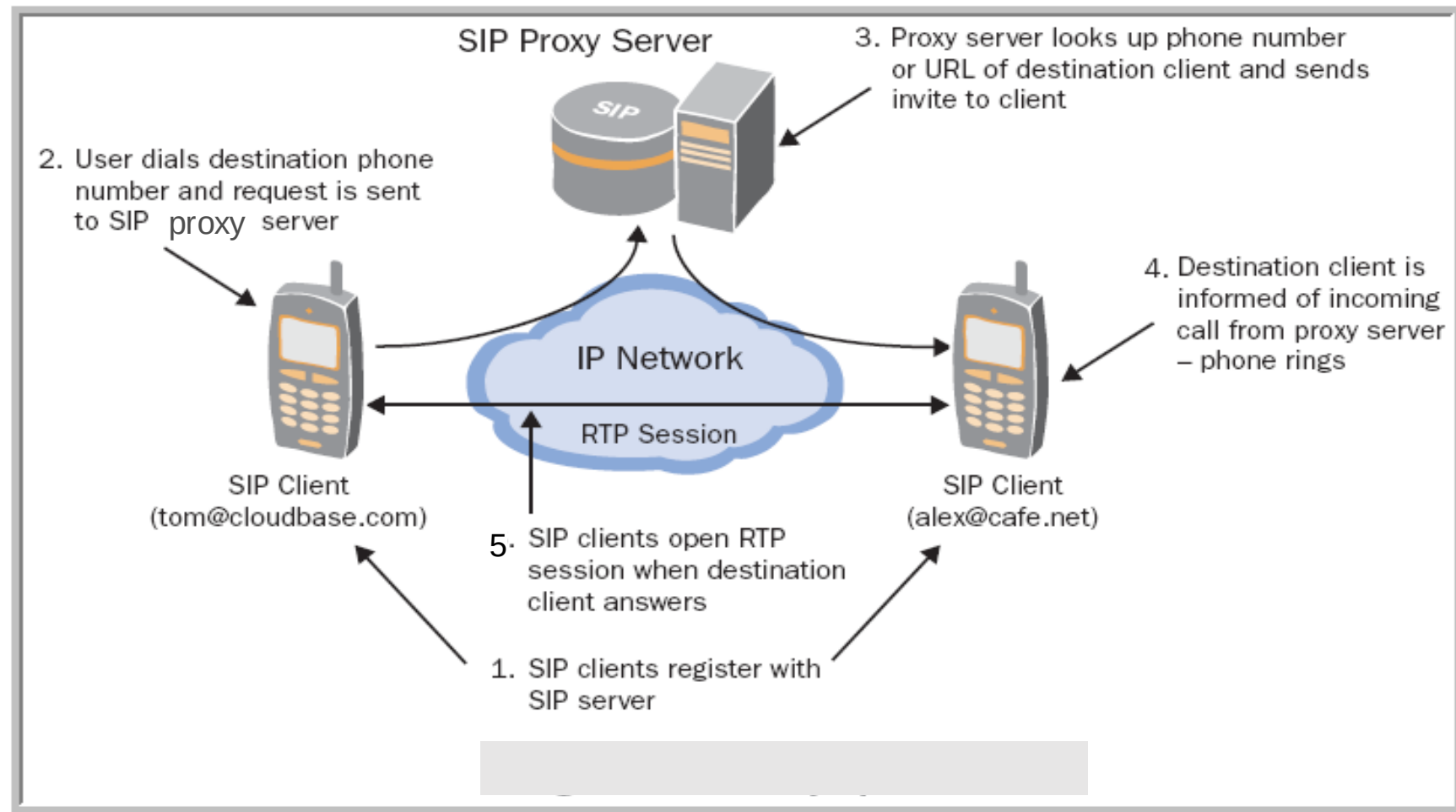


SIP Addressing

- The SIP address is identified by a SIP URL, in the format: user@host.
 - Examples of SIP URLs:
 - sip:skingsley@smu.edu
 - sip:skingsley@192.169.10.1
 - <sip:2147683960@smu.edu>
 - Can also use sips: for secure.



SIP Basic Call



SIP Requests

- INVITE: Initiates a call by inviting user to participate in session.
- ACK: Confirms that the client has received a final response to an INVITE request.
- BYE: Indicates termination of the call.
- CANCEL: Cancels a pending request.
- REGISTER: Registers the user agent.
- OPTIONS: Used to query the capabilities of a server.
- INFO: Used to carry out-of-bound information, such as DTMF digits.
- PRACK: Provisional response acknowledgement.
- REFER: Typically used to transfer a call to another party.
- SUBSCRIBE: Used to establish a subscription for the purpose of receiving notifications via NOTIFY.
- NOTIFY: used to convey information about the occurrence of a particular event.
- MESSAGE: Used to transport Instant Message (IM) information.
- UPDATE: Allows a client to update parameters of a session (such as codecs used).

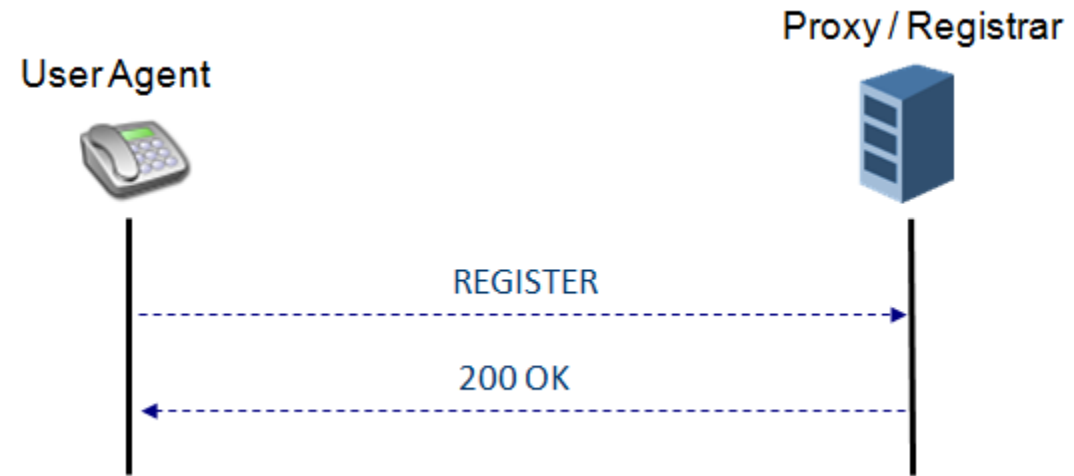


SIP Responses

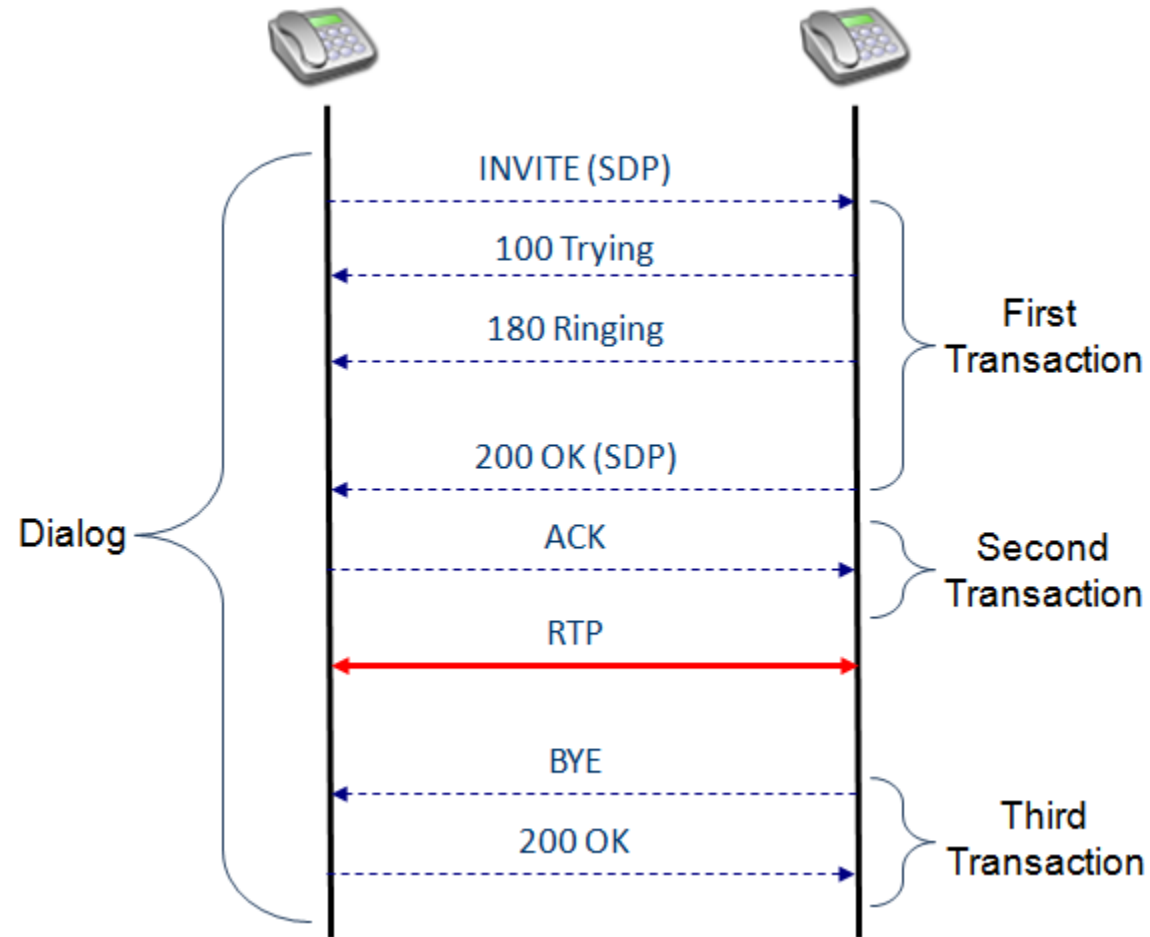
- Two types:
 - Provisional (informational)
 - Final
- Six classes:
 - Provisional (1xx): Indicates that the request was received and being processed.
 - 100 Trying
 - 180 Ringing
 - 183 Session Progress
 - Success (2xx)
 - 200 OK
 - Redirection (3xx)
 - Client Error (4xx)
 - Server Error (5xx)
 - Global Failure (6xx)



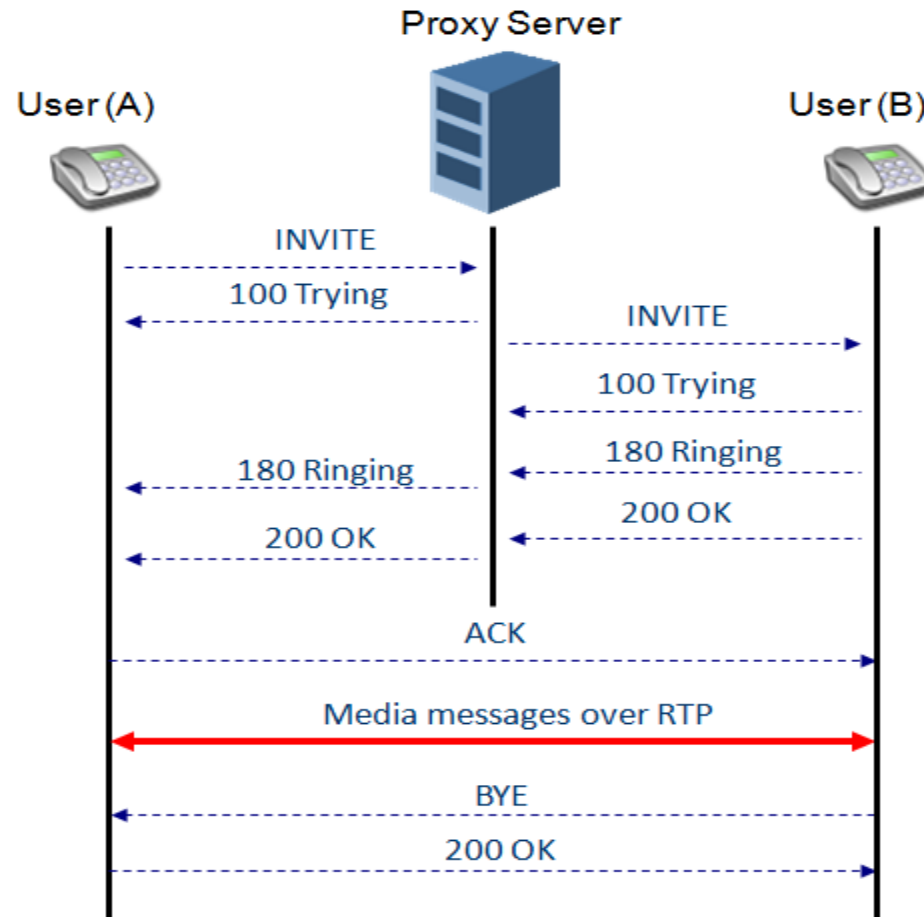
SIP Register



SIP Dialog Peer-to-Peer



SIP Dialog via Proxy Server



SIP Message Format

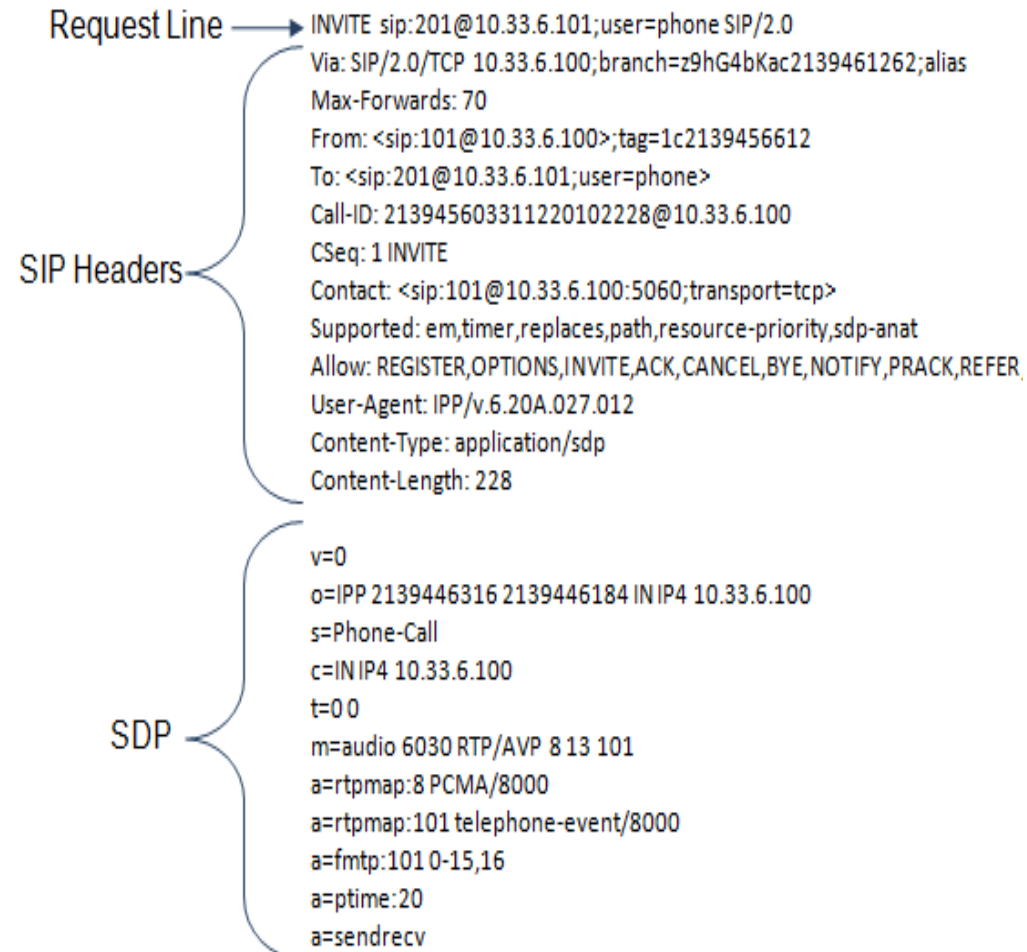
- Request Line
- Headers
- Body



SIP Message Request Line

- Request Line

- Mandatory
- Contains:
 - Method
 - INVITE
 - Request-URI
 - sip: 201@10.33.6.101
 - SIP version
 - 2.0



Summary/Takeaways

- Discuss the role of RTP in voice communications and show how its protocol characteristics are used to optimize Voice traffic on a network.
- Recognize various components required to deploy SIP, and how they are used for voice and video conferencing.

